GridMind

AI and Blockchain Integration in Smart Grids for Autonomous Energy Trading and Risk Optimization

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Research objective and market role

Application target: a price-taking BESS trader in Poland

Completed measured study: six Irish households under assumed tariffs

Community outcome: net cost and the distribution of savings

Polish preparation: physical scheduling and forecast timing checks

Polish market profitability and a daily risk constraint remain untested.

Experiments and their connections

Experiment	Comparison	Evidence
Allocation	Proportional, welfare LP, Nash	Savings and fairness
Battery control	Control and forecast interventions	Cost, feasibility and household outcomes
Settlement	Solidity and signed SQLite	Balance parity and rejection checks

New intervention: causal forecasts feed the same MPC and change realized cost.

Measured profiles and chronological evaluation

Data or split	Recorded coverage
Source	StoreNet Ireland, households H1–H6, 2020
Training	404 complete four-hour episodes, January–April
Validation	105 episodes, May
Evaluation	44 episodes, 1–14 June
Retained test intervals	704 of 1,344 intervals, 52.38%

Batteries and tariffs are assumptions. June dates had already been inspected in the allocation pilot.

Battery feasibility and causal observations

Constraint	Implementation
Capacity and power	2 kWh nominal, 2 kW AC per household
Operating bounds	0.2–1.8 kWh, 95% charge/discharge efficiency
Boundary condition	Every episode starts and finishes at 1 kWh
Available information	Previous net load, current SOC, clock and known tariff
Action protection	Power, SOC and terminal-reachability projection

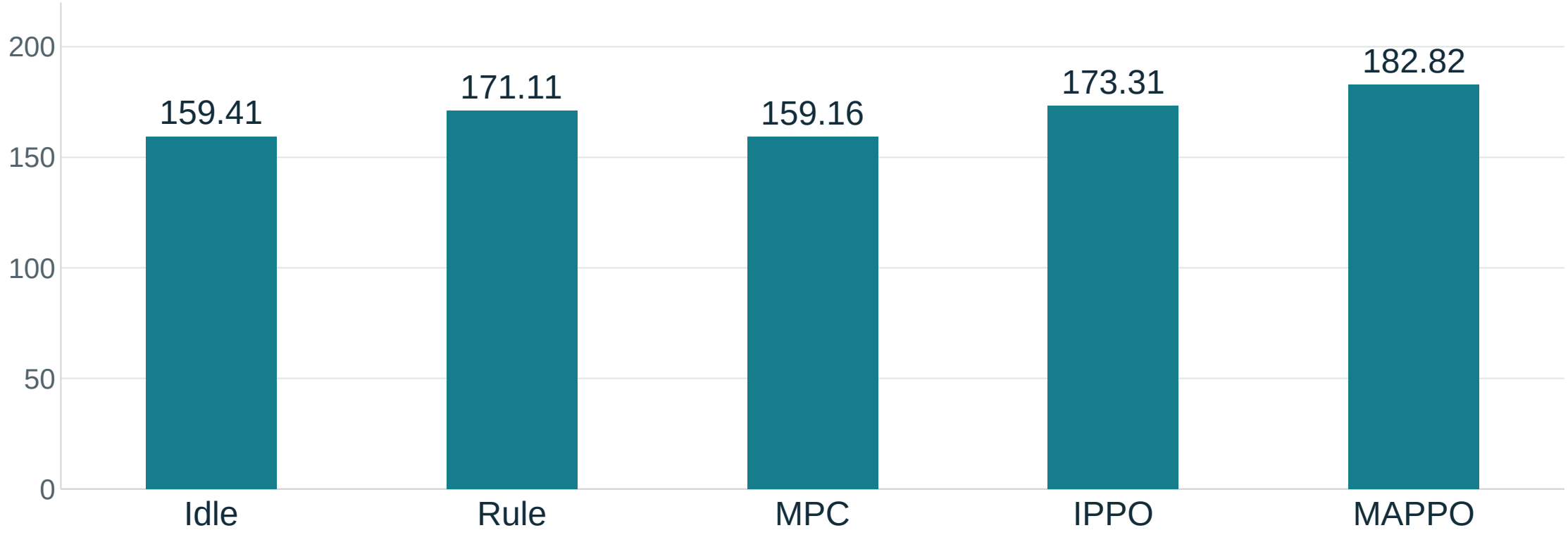
Terminal restoration has a cost. No controller receives free energy at an episode reset.

Learning comparison and training budget

Setting	IPPO	MAPPO
Actor at execution	Local, shared parameters	Local, shared parameters
Critic during training	Local observations	All household observations
Reward	Mean community incremental savings	Same reward
Random seeds	11, 22, 33, 44, 55	Same five seeds
Budget per seed	11,520 joint steps	11,520 joint steps

Original 60-update protocol. The follow-up varies actions, budget and reward.

Original control comparison at 60 updates



EUR over the same 44 episodes. IPPO/MAPPO bars show five-seed means. Seed SD: 19.51 / 22.42 EUR.

Efficiency and fairness can rank policies differently

Allocation method	Savings EUR	Mean daily normalized Jain
Welfare LP	26.0954	0.7741
Nash, 5% tolerance	25.8716	0.8187
Proportional	24.6755	0.8642

Heterogeneous-tariff allocation pilot. Its grid-only baseline differs from the battery experiment.

Both ledgers reproduce the same 64 settlements

Measurement	Observed result
Final balances	Exact match across seven accounts
Total accounting credit	6,000,000,000 micro-EUR units conserved
Solidity deployment	641,807 gas
Settlement gas	4,424,772 total, 68,917 median
Rejection and signature checks	10 of 10 passed

Local PyEVM chain and Ed25519-signed SQLite. No real currency or public-chain performance claim.

Verification and evidence limits

37 research tests cover physical, causal and data boundaries

70 saved checkpoints include 60 controlled follow-up runs

Maximum energy-balance residual remains below 1.4×10^{-15} kWh

Later evaluation adds 501 episodes across 168 days in the same six homes

No enforced daily CVaR constraint, feeder validation or demonstrated annual return on investment.

June follow-up findings

Discrete actions at 300 updates reproduce idle cost in all 10 runs

The fixed fairness penalty gives no consistent improvement

Ridge MPC costs 158.8618 EUR, with losses for two households

First-step forecast accuracy and control value rank differently

June remains exploratory. Backups G and H report the later temporal evaluation.

Conclusion

The prototype connects causal forecasts, MPC and physical accounting

Learning variants show no aggregate cost advantage over MPC

Community gains and correct ledgers do not guarantee individual benefit

Later profiles extend the evidence. Independent confirmation remains open

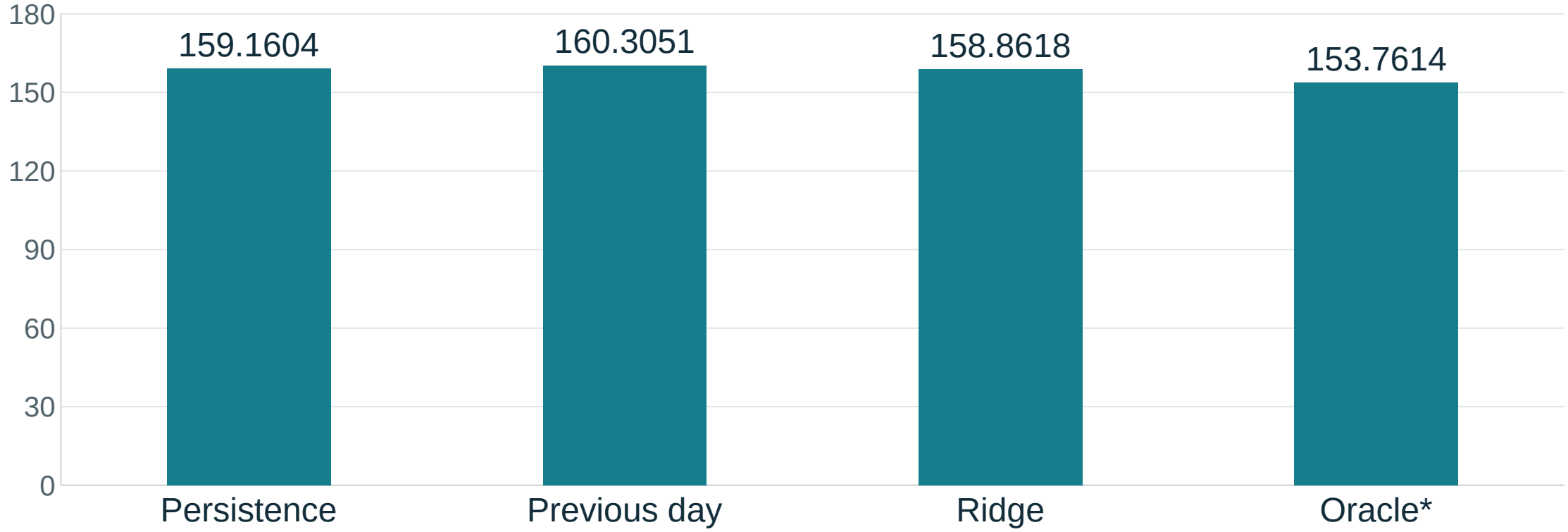
Questions

Backup A Original controller results

Controller	Mean cost EUR	Seed SD EUR	Saving vs idle EUR
Idle	159.4083	n/a	0.0000
Rule	171.1107	n/a	−11.7024
MPC	159.1604	n/a	0.2478
IPPO	173.3106	19.5128	−13.9023
MAPPO	182.8197	22.4156	−23.4114

SD describes variation among five training seeds, not independent households or days.

Backup B Forecasts change MPC cost



EUR over the same 44 episodes. Oracle uses future measurements and is non-causal.

Backup C Tariffs and economic boundary

Scenario input	Assumed value
Import before 07:00	0.20 EUR/kWh
Import 16:00–20:00	0.42 EUR/kWh
Other import periods	0.30 EUR/kWh
Export and local fee	0.08 and 0.02 EUR/kWh
Battery throughput cost	0.01 EUR per AC kWh

Historical measured profiles replay under scenario tariffs. No observed retail bill or market-price join.

Backup D June risk and household outcomes

CVaR95 averages the worst 5% of episode costs

44 episodes provide only 2.2 tail episode-equivalents

Ridge MPC saves in total while H1 and H5 lose relative to idle

A community saving does not establish individual rationality

No daily risk budget or learned fairness guarantee is enforced.

Backup E Evidence and demo route

Open the offline results viewer for controller and household outcomes

Inspect the 60-run ablation and four-arm forecast results

Use both original and follow-up evidence verification scripts

Saved-policy replay matches the recorded follow-up costs

The offline viewer displays recorded results. A fresh evaluation is a separate command.

Backup F Selected primary sources

Trivedi et al. (2024), StoreNet community dataset

Yu et al. (2021/2022), PPO in cooperative multi-agent games

Jain et al. (1984), resource-allocation fairness

Rockafellar and Uryasev (2000), conditional value at risk

Wüst and Gervais (2018), blockchain trust-model decision

Complete references and links appear in the manuscript and speaker notes.

Backup G Later period outcomes

Controller	Cost EUR	Saving vs idle EUR
Idle sharing	1947.4997	0.0000
Self-consumption rule	2065.3425	−117.8428
Persistence MPC	1946.6754	0.8243
Previous-day MPC	1966.2359	−18.7361
Ridge MPC	1939.3432	8.1565
Discrete IPPO 300, five-seed mean	1947.4997	0.0000

501 common episodes, July–December 2020. Frozen models, scenario tariffs.

Backup H Coverage and uncertainty

501 episodes across 168 days cover 45.38% of candidate blocks

Only 1 midnight and 11 morning blocks starting at 04:00 remain

Ridge minus persistence: −0.04364 EUR per retained source day

95% seven-day block interval: [−0.06889, −0.02127] EUR per day

Conditional exploratory inference. Prior access and incomplete coverage limit the claim.